Online Event Management System

https://github.com/aiyshwary/Online-Event-Management-System

SQL NoSQL Database Design Normalization

OVERVIEW

A database-driven system for managing events end-to-end, allowing customers to register, browse and book events with discounts, while giving administrators full control over event listings, users, payments and preferences.

IMPACT

Designed and implemented a normalized relational database and NoSQL schema for an Online Event Management System, covering customer registration, booking workflows, discount management, and payment processing.

TECHNICAL WALKTHROUGH

Online Event Management System – Project Explanation This project is a database-driven Online Event Management System designed to help users register, browse, and book events online, while allowing admins to manage events, users, discounts, and payments efficiently.

Problem it Solves
Managing events manually is time-consuming and error-prone—especially

when handling

i) Multiple event types

ii) User registrations

iii) Discounts and special preferences

iv) Payments and bookings

v) This system provides a centralized, structured database that automates all these processes.

vi) Key Actors

vii) User

viii) Registers using email and password

ix) Browses events

x) Books events with number of seats

xi) Applies discounts and special preferences

xii) Makes payment using multiple modes

xiii) Can update or cancel bookings

xiv) Admin

xv) Logs in with admin credentials

Adds, updates, or deletes

i) Events

ii) Discounts

iii) Special preferences

iv) User details

v) Manages pricing and discount rates

vi) Core Database Modules

vii) User

Stores registered user details

i) User ID, Email, Username, Password

ii) Users must register before booking any event.

iii) Event Details

Stores complete event information

i) Event name, type, date & time

ii) Duration and base price

iii) Each event belongs to a specific event category.

iv) Event Discount

Manages discount options

i) Premium account

ii) Group booking

iii) Coupon-based discounts

iv) Each discount has a rate reduced value.

v) Special Preference

Optional add-ons for users

i) VIP seats

ii) Backstage access

iii) Exclusive services

iv) These come with extra charges.

v) User Booking

Acts as the core linking table

i) Connects User, Event, Discount, and Special Preference

ii) Stores number of seats booked

iii) Each booking has a unique Booking ID

iv) This table enforces referential integrity using foreign keys.

v) Amount

Handles payment details

i) Payment ID

ii) Booking ID

iii) Mode of payment

iv) Admin

v) Stores admin login credentials and permissions.

vi) Database Design Highlights

vii) Fully normalized (3NF) to avoid redundancy

viii) Strong foreign key relationships

ix) Uses indexes for faster search

x) Supports views for business insights

xi) Important SQL Features Used

xii) Queries

xiii) Simple queries

xiv) Aggregate queries

xv) Join queries

xvi) Views

Examples

i) Total amount payable by each user (including discounts & extra

ii) charges)

iii) Users who registered but never booked

iv) Number of bookings per user

v) Events sorted by price for easy searching

vi) Indexes

vii) Faster event search by type & name

viii) Unique email constraint for users

ix) Faster lookup for bookings, discounts, and prices

x) Overall Summary

xi) This project implements a relational database–based Online Event Management System that efficiently handles user registrations, event bookings, discounts, special preferences, and payments with strong data integrity and optimized query performance.

KEY DETAILS

1. Designed a fully normalized relational schema (up to 3NF) with entities for Users, Events, Bookings, Payments, and Discounts.

2. Modeled equivalent NoSQL document structures for flexible event-type storage and preference handling.

3. Admin interface supports CRUD operations on events, user management, and discount configuration.

4. Customer portal enables account creation, event browsing, booking with preference selection, and payment tracking.

5. Schema documented with ER diagrams, normalization steps, and a detailed project report.